

Week 2 Lab: Diffusion Models (DDPM & DDIM)

Exercise 1

- In one or two sentences each, describe the *forward process* and the *reverse process* of a diffusion model. Which of the two contains learnable parameters?
- The lecture’s “denoise” block does not output a clean image directly; it contains a *noise predictor*. What does the noise predictor receive as input, and what does it output?
- The same network is used for all 1000 denoising steps. Why must the step index t be given to it as an input?

Exercise 2

The forward process adds noise one step at a time: $x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_t$, with independent $\varepsilon_t \sim \mathcal{N}(0, I)$.

- Show that two steps can be written as a single step, $x_2 = \sqrt{\bar{\alpha}_2} x_0 + \sqrt{1 - \bar{\alpha}_2} \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$. (Use: the sum of two independent zero-mean Gaussians with variances a and b is a zero-mean Gaussian with variance $a + b$.)
- State the general result for x_t and explain in words why it matters for *training*.

Exercise 3

DDPM uses a linear schedule with $\beta_1 = 10^{-4}$ rising to $\beta_{1000} = 0.02$. For this schedule, $\bar{\alpha}_{10} \approx 0.998$, $\bar{\alpha}_{100} \approx 0.897$, $\bar{\alpha}_{250} \approx 0.524$, $\bar{\alpha}_{500} \approx 0.079$ and $\bar{\alpha}_{1000} \approx 4 \times 10^{-5}$.

- For each t , give the weight of the clean image $\sqrt{\bar{\alpha}_t}$ and of the noise $\sqrt{1 - \bar{\alpha}_t}$. Sketch how an image x_t would look.
- The *signal-to-noise ratio* is $\text{SNR}_t = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$. Why should SNR_T be close to zero? What goes wrong at sampling time if it is not?

Exercise 4

Algorithm 1 (Training) of DDPM repeats: sample x_0 ; sample $t \sim \text{Uniform}\{1, \dots, T\}$; sample $\varepsilon \sim \mathcal{N}(0, I)$; take a gradient step on $\|\varepsilon - \varepsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, t)\|^2$.

- Identify the network input, the target and the loss in this line. How does this correspond to the loss computed in Lab 1A?
- The slides contrast training “ideally” (add noise step by step) with “in reality”. What is the practical difference, and why does it not change what the network learns?
- Why is t sampled uniformly at random in each iteration instead of looping over $t = 1, \dots, T$ for every image?

Exercise 5

DDPM is trained by maximising a lower bound (ELBO) on $\log P_\theta(x_0)$, which splits into three terms: (i) a reconstruction term $\mathbb{E}_q[\log P(x_0|x_1)]$; (ii) $\text{KL}(q(x_T|x_0) \| P(x_T))$; (iii) a sum over t of $\text{KL}(q(x_{t-1}|x_t, x_0) \| P_\theta(x_{t-1}|x_t))$.

- Why is maximising $\log P_\theta(x_0)$ directly “very difficult”?
- Which of the three terms depend on the network parameters? Explain the lecture’s remark that term (ii) has “nothing to do with the model”.
- Both distributions in term (iii) are Gaussians with (fixed) variances. Explain in words why minimising this KL reduces to making the network’s mean match the mean of $q(x_{t-1}|x_t, x_0)$, and why this in turn becomes the simple noise-prediction loss.

Exercise 6

DDIM reuses the U-Net trained for DDPM but samples with fewer steps. With $\eta = 0$ it is deterministic.

- (a) Why can DDIM use the *same* trained network as DDPM without retraining?
- (b) In the lab you compare 50 and 10 DDIM steps with DDPM (1000 steps). What do you expect to change in image quality and in sampling time?
- (c) What does “deterministic” mean here, and why does it make the sampler (approximately) invertible? Why is a DDPM sample not invertible in the same way?